

Why B1's quiet families fail

A four-expert diagnosis of the encoding-floor result (Plan 08 / B1). Retold in plain language.

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Four independent experts — a DSP engineer, an ML engineer, a volatile-memory engineer, and an ECG signal scientist — each analyzed the real data on the laptop, took a position, and named a falsifying test. This document is their discussion (Part 1) and their individual verdicts (Part 2), retold so a reader who does not live in this material can hold the whole thing in their head.

Part 1 — The discussion, retold

*The spine: we asked why five of seven workload families scored near zero. The uncomfortable answer the four converged on is that what looked like the method recognizing **workloads** is largely it recognizing the machine's idle **background** — and the workloads that genuinely write are the ones it rejects.*

THE SETUP, IN ONE PARAGRAPH

B1 turns memory into one "how busy" number per half-second (APF = fraction of pages changed), cuts it into 8-reading windows, and asks a bank of per-family autoencoders which family a window belongs to. It recognized the two write-heavy families (app 0.84, mem 0.75 on unseen workloads) and scored the five compute/read/thin families near zero. The question put to the experts: is that a too-short-trace problem, a too-small-window problem, or a too-poor-metric problem?

STORY 1 Three obvious explanations, all dead on arrival

Every easy answer failed on the laptop, with data in hand. **Window too small?** The DSP engineer sized it: 8 samples at 2 Hz spans 4 seconds and can only represent a level plus one slow ramp — but the signal decorrelates within one or two samples for **every** family, so there is no rhythm a bigger window could recover. **Traces too short?** Even at the maximal window (the whole trace) the quiet families stay one overlapping band. **They write nothing?** The ECG scientist's surrogate test found 100% of the quiet series are **structured**, not flat — there is a heartbeat. All three easy hypotheses ruled out before the debate even started.

STORY 2 The model was broken — and fixing it doesn't save you

The ML engineer found the autoencoder bank was pathologically degenerate: for the quiet families, every family's autoencoder rebuilds their near-flat windows equally well (error

~0.0001), so the "which family" decision is settled by fourth-decimal noise — a synthetic cpu window gets labeled thread. Swapping in a random forest on the *same* windows lifted cpu from 0.024 to 0.457 and thread from 0.010 to 0.568 — a 19x-to-59x jump from changing only the model. But the ECG scientist, running her own forest, noted the sober truth: that "0.4" does not beat the majority baseline (0.39). The forest crushes the broken autoencoder yet only reaches chance. **The model bug is real and fixable, but fixing it does not solve the problem.**

STORY 3 The reversal: it recognizes the floor, not the workload

One test settled it. We ran the forest recall *per workload* instead of per family, and it inverted everyone's guess:

workload	forest recall (unseen-workload)	what it was predicted to be
thread_lock_contention	0.833	predicted a wall
cpu_branch_random	0.725	predicted a wall (read-only)
thread_producer_consumer	0.714	—
cpu_hash_loop	0.678	predicted a wall (read-only)
thread_parallel_alloc	0.114	predicted re-representable
cpu_matrix_mult	0.011	predicted "the structured writer!"
io_direct_write / io_read	0.000 / 0.000	—

The forest recovers the workloads that barely write and rejects the ones that write the most. The reason is the test itself: to judge an unseen cpu workload, the model trains on the *other* cpu workloads. cpu is two floor-riders plus one real writer (matrix_mult). Hold out matrix_mult and the model has only floor-riders to learn from, so the one workload that actually writes looks nothing like "cpu" and is rejected. Hold out a floor-rider and its floor twin is in training, so it is recognized.

STORY 4 So "family" was never a real question here

Put the reversal together with the memory engineer's floor finding — the workloads designed to write nothing still show ~118 changed pages every half-second, the operating system's own background (timekeeping, scheduler, per-CPU counters) — and the picture is clear. **The model is largely recognizing the OS background, not the workload.** Each quiet family is mostly floor-riders sharing an idle signature, with a rare real writer as the odd one out. The classifier

learns the majority — the floor — which is why it "recognizes" the floor-riders and rejects the writers. The target itself is incoherent: "family" averages a writer with near-silent members that share nothing but the machine's idle hum. This also throws a shadow on app and mem: their 0.84 / 0.75 may be plain *level* recognition (loud vs quiet), not behavior recognition.

STORY 5 What is actually true, and what to do about it

- **Not silence, but not family-signal either.** The quiet workloads write and are structured, but their distinguishing information is not in write-*breadth* (APF) or write-*magnitude* (proven to the ceiling on the laptop) — it is in *where* and *what* they write: the spatial and directional channels APF averaged away.
- **A genuine wall exists.** The pure readers and spinners (`io_read_cache_hit`, `lock_contention` as a `futex spin`) do their work in reads, disk, registers, and the scheduler — places a memory-write diff physically cannot see. No transform invents a signal that never wrote.
- **The founding lens was switched off.** At the campaign's speed setting, the phase / FFT-shift channels the whole idea centered on were literally not computed (they emit zeros). Testing them needs a few targeted re-captures.

The shape of it. B1's "success" is substantially the model recognizing the machine's idle floor and the loud-vs-quiet level, not workload behavior; the workloads that truly write are the ones the family model rejects; and the honest signal — where any exists — is spatial and per-workload, not per-family. What is urgent: one *idle-baseline* capture (does idle equal the floor-riders? — the clean proof the "recovery" is environment, not workload) and one *spatial* pass over already-captured per-page data (do the real writers have distinct write-location signatures?). Both are cheap and specific; neither needs the full corpus.

Part 2 — The four verdicts, retold

Each expert reached an independent position on the same data. Here is each one's finding in plain language, with the number that carries it and their one-line verdict.

THE DSP ENGINEER It's not time, it's space

He did the arithmetic first: an 8-sample window at two-per-second can resolve almost nothing — a level and one slow ramp — and the signal decorrelates within one or two samples anyway, so no window length recovers a rhythm that isn't there. Then he showed the discriminator that *does* exist is being thrown away: the average bits-changed-per-page differs 5x across the quiet families (cpu 0.137, io 0.093, thread 0.027), a real magnitude signal that lives in the per-page data and is destroyed the moment you average 262,144 pages into one number.

*Verdict: the cause is the metric's **spatial averaging**, not the window or the trace length. The quiet families write (~130–150 pages a snapshot) — the information is still in the per-page field, just averaged out. Falsify me by computing spatial/magnitude features from the already-captured per-page data and re-running the hard split.*

THE ML ENGINEER The autoencoder was the bug, but not the whole story

He proved the autoencoder bank collapses the quiet families *by construction*: their windows are so flat that every family's autoencoder reconstructs them equally, so the winner is noise. A plain random forest on the identical windows and splits lifted cpu 19x and thread 59x — the signal was in the window, the model was discarding it. He was careful about the ceiling: the honest gains survive the leave-one-workload-out split, but the thin families (io with two workloads, cache, sandbox) are a *data* limit no model fixes.

Verdict: the dominant fixable fault is the model/scoring rule; the metric limits only io; the thin families are limited by too few workloads. Falsify me by running a forest on the raw windows — if it also flatlines, it is representation or data, not the model. (It did not flatline; the position survived, but the follow-up per-workload test showed the "recovery" is floor-recognition, not behavior — a caveat he flagged as his lens's blind spot.)

THE VOLATILE-MEMORY ENGINEER Below the noise floor of the machine

His was the reality check. The two workloads *designed* to write nothing — a cache-resident hash loop, a register-accumulating branch test — still register ~118 changed pages a snapshot, which is the operating system's own background. Every quiet workload sits within a few dozen pages of that floor: **the workload signal is smaller than the OS noise it rides on**. Reading the actual workload source, he split the families by workload: one real structured writer (`matrix_mult`, a 512-page contiguous matrix), one aliasing case (`parallel_alloc`, whose recycled pages are rewritten thousands of times but counted once per snapshot), and the rest genuine walls whose work is reads, direct-to-disk writes, or scheduler state a RAM diff cannot see.

Verdict: mostly an observability wall, per workload, not per family — the quiet families' identity lives in channels this instrument does not record. Falsify me by capturing an idle guest: if the near-silent workloads are statistically identical to idle, no representation of their (absent) writes can classify them.

THE ECG SIGNAL SCIENTIST A heartbeat, but not a diagnostic one

She held the founding analogy to account with numbers. A surrogate test proved none of the quiet families is a flatline — 100% are structured, all autocorrelated; there *is* a beat. But she then swept every feature she could build from the laptop's two scalars — shape, spectral, longer windows, magnitude, the two-channel pairing — and none beat chance. The beat exists but looks the same in this lead. Crucially, she caught that the phase and FFT-shift channels the whole idea rests on were *switched off* at the campaign's speed setting, and that each quiet family is not acoustically one family — it pairs a writer with a near-silent member, in most of all.

Verdict: the founding intuition is vindicated but bounded — the real writers (`matrix_mult`, `producer_consumer`, `parallel_alloc`, `io_direct_write`) are re-representable through the spatial and directional leads that were averaged away or turned off; the pure readers and spinners (`io_read_cache_hit` above all) are true asystole.

Restore the leads for the former; expect nothing from the latter. Falsify by computing the spatial write-location signature from the existing per-page data.

Where they agree, in one place

- **Ruled out (all four):** window size and trace length are not the cause.
- **Agreed:** the quiet families write and are structured, but sit at/below the OS idle floor; their discriminative signal — where any exists — is spatial/directional, not temporal or breadth-based.

- **The reversal all four now accept:** the family "recovery" is largely floor/level/environment recognition, and the real writers are the workloads the family model rejects — so "family" is not a coherent target for the quiet workloads.
- **The genuine wall:** pure readers/spinners are unobservable to a write diff; no metric or model recovers them.

The two cheap tests that would confirm all of this (specific, not the full corpus):

1. Capture an *idle* guest and compare it to the near-silent workloads — proves the "recovery" is the OS floor, not the workload.
2. Compute *spatial* write-location features from the already-captured per-page data and re-run the hard split — proves whether the real writers (matrix_mult, producer_consumer, parallel_alloc, io_direct_write) carry a distinct signature the scalar threw away.

Sources: independent Round-1 reports from the dsp-engineer, ml-engineer, volatile-memory-expert, and ecg-signal-scientist agents; the orchestrator's per-workload forest test on ~/b1_data/b1_windows.npz. Companion:

B1_RESULT_EXPLAINED.md, plan08_overview.html, EXPERIMENT_B1_ENCODING_FLOOR.md. All figures computed on the laptop copy of the B1 windows; server-side confirmations (idle baseline, per-page spatial pass) are proposed, not yet run.